Senthil Kumar Dorairaj — Applied Gen AI & Agentic AI Engineering Portfolio

Applied Generative AI | Agentic AI Systems | LangGraph | LangChain | RAG | Compliance Engineering

Johns Hopkins University — Applied Generative AI Program & Advanced Agentic AI Program

1. Automated Mortgage Underwriting System

Program: *Advanced Agentic AI — Johns Hopkins University*

Tech: LangGraph, LangChain, GPT-4o-mini, ChromaDB, Python, RAG, PII Sanitization, Bias Detection

Overview: An enterprise-grade, agentic AI multi-agent workflow that automates mortgage underwriting with deterministic tools, compliance safeguards, and auditable reasoning chains.

Key Capabilities:

- Hierarchical multi-agent architecture: Credit, Income, Asset, Collateral, Supervisor, Critic, Decision agents
- Deterministic underwriting tools: DTI, LTV, reserves, housing ratio
- RAG policy retrieval for regulatory alignment (Fair Lending Act)
- PII sanitization + bias detection for compliance
- HITL workflows for high-risk decisions
- LangGraph state machine for deterministic orchestration

Impact: Reduced underwriting time from days to hours, improved consistency, and enabled scalable, compliant decision automation.

2. Autonomous Financial Research Analyst

Program: *Advanced Agentic AI — Johns Hopkins University*

Tech: LangGraph, LangChain, OpenAI API, RAG, yfinance, Tavily, Sentiment Analysis

Overview: A goal-oriented autonomous agent that performs financial research, analyzes market sentiment, retrieves proprietary insights, and generates investment-aligned reports.

Key Capabilities:

- Five specialized tools: stock price, stock history, news search, sentiment analysis, private RAG database
- Proactive, reactive, and autonomous agent behaviors
- Full RAG pipeline for grounded insights with citations
- Multi-company ranking based on financial performance + AI innovation
- Synergistic tool usage demonstrating holistic analysis

Impact: Delivered timely, data-driven insights and multi-company evaluations across financial and innovation dimensions.

3. AI for Research Proposal Automation

Program: *Applied Generative AI — Johns Hopkins University*

Tech: LangChain, LangGraph, FAISS, RAG, PDF Parsing, LLM-as-Judge

Overview: An AI system that transforms raw scientific papers into fully evaluated, NOFO-aligned NIH research proposals.

Key Capabilities:

- Topic extraction from NOFO documents
- FAISS-based semantic relevance ranking of research papers
- Generation of five fundable research ideas with citations
- Structured proposal blueprint creation
- LLM-as-Judge scoring across Innovation, Significance, Approach, Expertise

Impact: Automated the full proposal lifecycle, reducing manual effort and improving consistency and alignment with funding criteria.

4. AI Email Secretary — Prioritization & Response Automation

Program: *Applied Generative AI — Johns Hopkins University*

Tech: OpenAI API, Python, LLM-as-Judge, Semantic Categorization, Yesterbox Workflow

Overview: An AI assistant that prioritizes emails, summarizes inbox activity, drafts responses, and generates a daily executive communication report.

Key Capabilities:

- Yesterbox-based inbox summarization
- Six-category LLM-driven email classification
- Executive Dashboard summarizing inbox health
- Detailed urgent + deadline-driven summaries with next steps
- AI-generated first-response drafts for critical emails
- LLM-as-Judge evaluation of clarity, relevance, actionability

Impact: Improved communication efficiency, reduced manual email load, and enhanced decision-making clarity for high-volume inboxes.

GitHub Portfolio

All projects available at: https://github.com/senthildorairaj